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SURVEY

A Systematic Literature Review on AI-Based Recommendation Systems and Their Ethical Considerations

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ABSTRACT With the rise of social media, individuals face challenges in decision-making due to the abundance of options available. Recommender Systems (RSs) leverage Artificial Intelligence (AI) to provide users with personalized suggestions aligned with their preferences and interests. This study presents a systematic review of AI-based Recommender Systems, focusing on recent advancements and primary studies published between 2019 and 2024. While several review papers have addressed various aspects of RSs, the rapid evolution of AI techniques necessitates an updated review to capture the latest trends and innovations. We systematically gathered data from five major databases: IEEE, Springer, Science Direct, ACM, and Wiley. Through the PRISMA methodology, we selected 85 relevant studies. Our analysis addresses several key research questions: the types of datasets and data sources used, major application fields, prevalent machine learning and AI techniques, overall research productivity, and the limitations and future trends in AI-based RSs. Our findings indicate that advanced AI techniques, particularly those incorporating deep learning with multiple hidden layers and transformer models like BERT, significantly enhance the accuracy and effectiveness of Recommender Systems. Furthermore, we observed a trend towards integrating contextual and real-time data to improve recommendation relevance. Additionally, we discuss ethical considerations such as privacy, data security, bias, and transparency, emphasizing the need for responsible AI development to ensure fair and equitable recommendations. These insights can guide future research and development efforts in the field.

INDEX TERMS Artificial intelligence, recommender systems, systematic literature review, ethical considering of RS.

I. INTRODUCTION

Making a correct decision is a challenging task, especially when faced with a broad spectrum of options. The sheer volume of choices makes it difficult for individuals to be satisfied with a single option. This phenomenon is called 'choice overload.' Due to the large number of choices, individuals try to weigh each option based on its pros and cons and consider their potential consequences, thus making the decision more difficult. Consequently, users need a decision-making assistant that is effortlessly automatic and fulfills their needs [1].

Recommendation Systems (RSs), or recommendation engines, are one of the solutions that help users manage choice overload. RSs are important tools that enable individuals to make choices aligned with their needs and preferences. These systems use algorithms and user data to provide personalized recommendations, helping users navigate through a large number of choices [2]. Thus, RSs provide suggestions that are aligned with users' specific interests by taking into

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account their behavior, interactions, and preferences. This ability to provide personalized recommendations not only improves the user experience but also tends to decrease choice overload [3]. RSs serve as valuable tools in optimizing user interaction by providing suggestions that align with users' interests. These suggestions can pertain to movies, products, drugs, articles, books, and other types of content [4].

With the advancement of technology, Artificial Intelligence has experienced rapid development, particularly in attention and transformer models. These models consist of multiple layers, each dedicated to extracting more complex information from the given text [5]. In these layered models, information is processed in each successive layer, refining the representation of input before passing it to the next layer. These processing layers allow the model to learn the complex and intricate patterns found in the input data, thus enabling it to perform numerous tasks, such as natural language processing (NLP), sentiment analysis, image processing, and more [6].

Recent advancements in attention and transformer models have significantly impacted the performance of Recommender Systems (RSs) [7]. These models have introduced several key improvements: **Enhanced Contextual Understanding:** Transformers, through their attention mechanisms, enable RSs to better understand and capture complex user-item interactions. By weighing the importance of different elements within the user data, these models can more accurately identify user preferences and provide personalized recommendations [8]. **Improved Handling of Sequential Data:** Attention mechanisms excel at processing sequential data, which is crucial in RSs for capturing temporal patterns in user behavior. This ability allows for more accurate predictions of future user interactions based on past behavior [9]. **Scalability and Efficiency:** Transformer models, such as BERT and its variants, are highly scalable and can handle large datasets efficiently. This scalability is particularly beneficial for RSs operating in domains with vast amounts of data, ensuring timely and relevant recommendations. **Versatility in Application:** The flexibility of attention mechanisms enables their application in various types of RSs, including collaborative, content-based, and hybrid systems. This versatility allows for the integration of diverse data sources, enhancing the overall robustness and effectiveness of the recommendations. **Improved Personalization:** By leveraging the power of attention mechanisms, RSs can offer more personalized and contextually relevant recommendations. This results in a better user experience, as the recommendations align more closely with individual user preferences and needs.

There exist different types of Recommender Systems since they became popular in the 1990s [10].

1) **Collaborative Filtering**: This is the most widely used type of RSs. It works based on similarity, which can be calculated in different ways:
   - **User-Based**: It finds similarity based on the ratings provided by similar users.
   - **Item-Based**: It finds similarity based on the ratings of already rated items and the targeted one.
2) **Content-Based Filtering**: This type of RS generates recommendations based on the profiles of users.
3) **Hybrid Filtering**: This approach combines different types of RSs to overcome the limitations of other techniques.

Recently, research and publications related to AI-based Recommender Systems have increased significantly. Numerous surveys have reviewed AI-based RSs, providing excellent summaries of high-quality research papers. However, no study has thoroughly explored advanced AI techniques for Recommender Systems. As a consequence, a comprehensive study and review of primary studies related to AI-based RSs is needed to describe their advantages, limitations, and possible future trends. This will help guide new research in developing AI-based RSs.

A. CONTRIBUTIONS

The manuscript provides a systematic literature review on recommendation systems, offering several unique contributions that extend beyond a mere summary of existing literature. Firstly, it identifies and categorizes emerging trends and novel methodologies in the development of recommendation systems, highlighting the evolution of techniques over time. Secondly, the review uncovers critical gaps and challenges in current research, offering a roadmap for future investigations. Thirdly, it provides a comparative analysis of various recommendation algorithms, assessing their performance and applicability across different domains such as healthcare and education. Lastly, the manuscript discusses the ethical implications and societal impacts of recommendation systems, offering insights into best practices for designing fair and transparent systems. These contributions collectively provide a comprehensive and nuanced understanding of the current landscape and future directions of recommendation systems research. Over the past decade, numerous research groups have tackled the aforementioned issues from various angles, offering intriguing solutions and insights. In this systematic literature review, we will try to give the reader an overview of the main approaches of Artificial Intelligence used in Recommender Systems. More in detail, we will give an introduction of the taxonomy used in Recommender System research. After the taxonomy description, we will discuss the datasets or data sources which are used in research related to Recommender Systems. Moreover, we will discuss the main approaches and algorithms of machine learning, deep learning and artificial intelligence used in research related to Recommender Systems. Furthermore, we will describe the evaluation matrices which are used to evaluate the performance of the Recommender Systems. More in detail, we will describe which application areas or real life scenarios are taking advantages of Recommender Systems. Moreover, we will provide the list of top publication venues, the year wise trend and the active countries in the research related to Recommender Systems. Furthermore, we will

discuss what are the existing challenges and future research gaps in the field of Recommender Systems.

Hence, this study tends to identify and explore the current research work in the field of AI-based Recommender Systems. The established research gaps and potential research directions are the expected outcomes.

The objectives of this study are:

- To identify the most widely used datasets and sources.
- To identify the techniques, algorithms, and evaluation methods used in research related to Recommendation Systems.
- To identify the advanced AI algorithms used in this field.
- To identify the main application areas that utilize AI-based recommendation systems.
- To assess the overall research productivity in this field.
- To identify existing limitations and future research directions.

The rest of the paper is structured as follows: Section II reviews related work, Section III explains the methodology, Section IV presents the results and discussion, Section V addresses ethical considerations, and Section VI concludes the paper

II. RELATED WORK

Recommender Systems (RSs) are developed to assist individuals in finding content based on their preferences and interests from a large pool of available options. RSs are software applications that provide users with recommendations for various items such as movies, songs, careers, jobs, medicines, and education [6].

In recent years, several studies have been conducted to review and survey classical RSs. A notable early work is by Dehdarirad et al. [6], who conducted a systematic review to investigate RSs that suggest scholarly publication venues to researchers. Choosing an appropriate publication venue is a crucial step in the publication of research papers, and new researchers often face confusion in selecting the right venue due to the variety of topics across academic disciplines.

In [11], Da'u and Salim et al. presented a systematic literature review (SLR) of recommendation systems based on deep learning techniques. They investigate emerging trends and techniques for addressing unresolved issues, as well as exploring open issues in deep learning-based RSs and providing future directions for new researchers. They analyzed and compared primary studies based on application domains, datasets, and different metrics used for evaluating the performance of deep learning-based RS approaches.

Similarly, in [1], Alrashidi et al. conducted a systematic literature review to identify, summarize, and assess the state-of-the-art in social recommendation systems using deep learning algorithms. Their review of 32 primary studies revealed potential gaps in the field of social recommendation based on deep learning. They also presented several social application domains, datasets, and evaluation metrics used in various social recommendations.

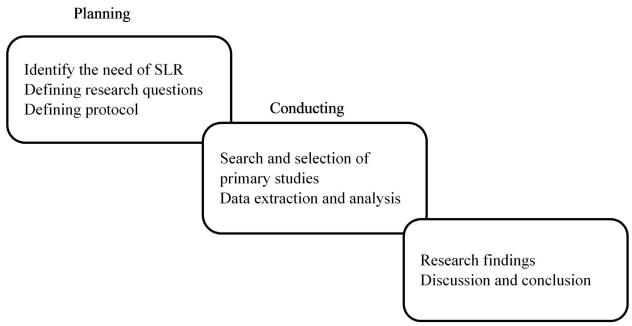

FIGURE 1. Research methodology.

In [5], Farina et al. conducted a systematic literature review (SLR) to explore techniques used in Metrics Recommender Systems for software engineers. Their focus was on datasets, algorithms, and recommendations in the SLR. They investigated the main preprocessing steps in metric Recommender Systems, as well as optimization strategies for such steps, identifying the most promising approaches for metric Recommender Systems.

Similarly, in [10], Idrissi and Zellou conducted an SLR to discuss proposed solutions for mitigating sparsity issues in collaborative filtering. They also investigated the types of side information necessary to solve sparsity problems and disclosed criteria for improving the accuracy of recommendation systems. For each selected paper, they examined evaluation metrics, datasets used, and sparsity/density levels, discussing the evaluation results.

In [3], Masruroh et al. conducted a systematic literature review (SLR) to assess the topics and methods applied in predictive analysis of educational data mining. This review served as an assessment process, identifying and interpreting existing studies to answer research questions. Through the analysis of selected studies, the results revealed that predictive data mining for academics currently focuses on eight topics, namely classification, Recommender Systems, educational data mining, knowledge discovery databases, student prediction, academic performance, and predictive analysis. Twenty-one different methods were applied to data mining prediction for academics, with five methods being most frequently used: K-Nearest Neighbor (K-NN), Support Vector Machine (SVM), Decision Tree (DT), Naïve Bayes (NB), and Random Forest (RF).

Silva et al. [4] conducted a systematic literature review (SLR) to explore the main trends and fields of educational recommendation systems. They selected primary studies from the past six years, focusing on those studies within the educational domain. The review investigated how educational Recommender Systems produce recommendations and how they are evaluated. This SLR provides a three-axis-based analysis centered on systems' underlying techniques, input data, and results presentation. Furthermore, it highlights future trends and limitations of studies in the field of educational recommendation systems.

TABLE 1. Research questions.

RQ #	Research Question	Motivation
1	Which data source/dataset is most frequently used in recommendation systems	To identify the data sources in this filed.
2	What are the techniques and algorithms and evaluation methods used in the research related to Recommendation Systems?	To compare the most used techniques, tools, algorithms and their effects on recommendation systems, as well as compare their evaluation strategies.
3	What are the main artificial intelligence algorithms used in this field.	To find the contribution of Artificial Intelligence in the field of recommendation systems?
4	What are the main application fields of the recommendation systems?	To identify the main fields where recommendation systems are widely used.
5	What is the overall productivity in the field of recommendation systems?	To analyze the year-wise trends, publication venues, active countries and contribution type.
6	What are the existing challenges and future research opportunities of the studies?	To identify the gaps in existing literature.

TABLE 2. Electronic databases and initial results.

ID	Databases	URL	No. of Studies
ED1	IEEE	http://ieeexplore.ieee.org/	204
ED2	Springer	http://link.springer.com/	174
ED3	ACM	http://dl.acm.org/	93
ED4	Science Direct	http://sciencedirect.com/	118
ED5	Wiley	http://onlinelibrary.wiley.com/	64

III. RESEARCH METHODOLOGY

Well-scheduled planning is crucial to ensure that only relevant studies appear in the literature review, aligning with the specific topic. The primary focus of the systematic literature review (SLR) is to identify, evaluate, and interpret existing literature related to the specific field. We adopted the guidelines of Kitchenham for conducting our SLR, adhering to a systematic and unbiased process. SLR comprises three phases, as illustrated in Figure 1:

Research Questions: The aim of this SLR is to identify what is in the state of art in the field of recommendation systems. The primary objectives of this SLR have divided into six questions as follows in Table 1:

Electronic Databases: We utilized five databases to collect studies published between 2019 and 2024. These databases are widely regarded as the best sources in the field of computer science. The number of studies found in each database is provided in Table 2:

Search Terms:

Relevant primary studies from scientific databases were collected by generating a query, as outlined in Table 3. The query includes relevant keywords selected to enhance the searching and filtering process.

Population: (Recommendation OR Recommender) AND (Systems OR System OR Engine)

Intervention: Artificial Intelligence OR AI.

TABLE 3. Search terms on each database.

Database Name	Search String
IEEE	((Recommendation OR recommender) AND systems) AND (Artificial Intelligence OR AI OR Machine Learning)
Springer	("Recommendation" OR "recommender") AND "systems" AND ("Artificial Intelligence" OR "AI" OR "Machine Learning")
ACM	{ Title:("recommender system") AND Title:(based on "artificial intelligence" OR "machine learning") }
Science Direct	"Recommender System" AND "artificial intelligence OR "machine learning""
Wiley	""recommender system"" in Title and "based on "artificial intelligence" OR "machine learning"" anywhere

Selection of relevant studies:

The PRISMA process was followed to select relevant studies, as outlined by Kitchenham, which includes four phases. The first phase, identification, involved collecting primary studies from databases and removing duplicates. The second phase, screening, assessed relevancy based on title and abstract, excluding irrelevant studies. Eligibility, the third phase, compared full-text articles with exclusion, inclusion, and quality criteria. The eligible articles were selected in the inclusion phase of the PRISMA process. The PRISMA flow is depicted in Figure 2. According to the PRISMA flow, 653 studies were obtained from the five databases. After duplicate removal, 521 primary studies remained. Subsequently, 220 studies were excluded based on title, resulting in 301 primary studies. Following abstract review, 146 primary studies were discarded, leaving 155 full-text articles for further consideration. After applying a data extraction form, 74 additional primary studies were discarded, resulting in 81 primary studies. Both backward and forward snowballing were then performed, yielding an additional 85 primary studies for review. Backward snowballing involved reviewing the reference lists of these selected studies to identify additional relevant papers that were cited by the included studies. This method helped us uncover earlier research that may not have appeared in our original database search but is pertinent to our review. Forward snowballing was conducted by examining the papers that cited the selected studies. By analyzing who has referenced our included studies, we identified newer research that has built upon or extended these foundational works. This dual approach allowed us to comprehensively capture the breadth of research related to AI-based Recommender Systems, ensuring that our review covered both seminal and recent advancements in the field.

Inclusion and exclusion criteria:

The inclusion and exclusion criteria were applied during the second (screening) and third (eligibility) phases of the PRISMA process when selecting primary studies for review. The inclusion and exclusion criteria are outlined in Table 4 and Table 5, respectively.

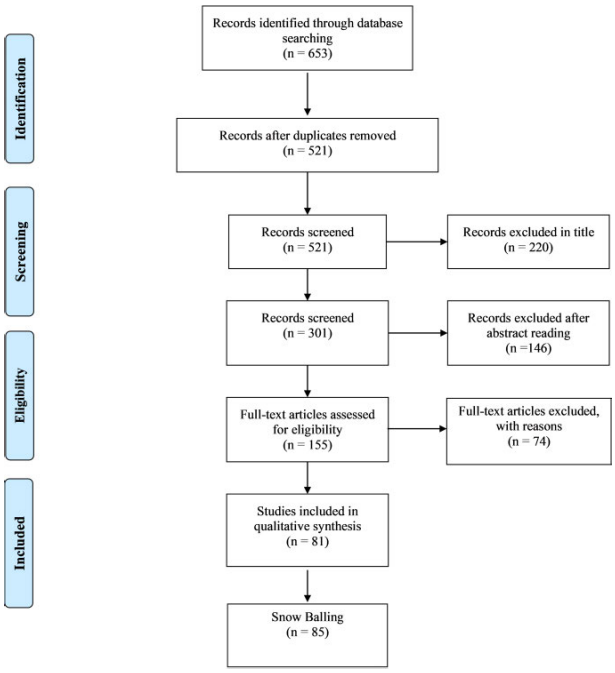

FIGURE 2. Prisma flow.

TABLE 4. Inclusion criteria.

IC#	Description
IC1	Peer Reviewed articles
IC2	Articles purely related to Machine Learning and Artificial Intelligence.
IC3	Only updated version of the same theme articles by same set of authors.
IC4	Articles published in 2019-2024 only
IC5	Full text available articles

TABLE 5. Exclusion criteria.

EC#	Description
EC1	Studies published in languages other than English
EC2	Studies not related to context of the research.
EC3	Editorials, short papers, posters, extended abstracts, review papers, blogs or Wikipedia pages should not be included
EC4	Articles providing unclear results and findings
EC5	Early cite articles will be excluded.

Potential Limitations and Biases in Inclusion and Exclusion Criteria: There were certain limitations in exclusion and inclusion criteria, which are as follows:

1) Scope of Databases: The review was conducted using five major databases: IEEE, Springer, Science Direct, ACM, and Wiley. While these databases are comprehensive, they may not cover all relevant studies, particularly those published in niche or less widely indexed journals. This limitation could result in the exclusion of potentially valuable research from other databases or sources.
2) Publication Bias: Our criteria primarily focused on published studies, which might introduce publication bias. Studies with negative or inconclusive results, which may not have been published, were not included. This could skew the review towards studies that report positive findings.
3) Language and Access Constraints: The review was limited to studies published in English due to language constraints. This exclusion might have omitted relevant research published in other languages. Additionally, access restrictions to full-text articles could have led to the exclusion of studies where full information was not available.
4) Date Range Limitation: The review focused on studies published between 2019 and 2024. While this timeframe ensures the inclusion of recent advancements, it might exclude significant earlier work that could provide valuable context or foundational knowledge.
5) Criteria Specificity: The inclusion criteria were designed to capture studies relevant to AI-based recommender systems with a focus on sentiment analysis. This specificity might unintentionally exclude studies that, while relevant, did not explicitly match all the predefined criteria.

Mitigation Efforts: To mitigate these limitations and biases, we employed a comprehensive search strategy, including snowballing to capture additional relevant studies and conducted a rigorous review process involving multiple reviewers to enhance the inclusiveness and accuracy of the study selection. We acknowledge these potential biases and limitations in our discussion to provide a transparent account of the review's scope and findings.

Handling 'Unclear' Studies:

1) Initial Assessment: During the data extraction phase, studies marked as 'Unclear' were initially flagged when there was insufficient or ambiguous information in the study's abstract or main text regarding key aspects relevant to our review.
2) Re-evaluation Process: These flagged studies were subjected to a thorough re-evaluation. This involved a secondary review by at least two independent reviewers who re-assessed the full text of each study to clarify the unclear aspects.
3) Consensus Meeting: If discrepancies or unresolved issues remained after the initial re-evaluation, the reviewers convened to discuss and reach a consensus. During this meeting, the reviewers reviewed the study's methodology and results in detail to ensure accurate classification.
4) Consultation with Experts: In cases where clarification was still needed, we consulted subject matter experts or referred to additional resources and supplementary materials provided by the study authors, if available.
5) Final Decision: After these steps, studies that could not be clarified were excluded from the final dataset.

We documented the reasons for exclusion to maintain transparency in our review process.

Quality assessment criteria: The purpose of the quality assessment criteria is to select high-quality studies for the systematic literature review (SLR). It is applied during the second and third phases of the PRISMA process. Only studies that pass the quality checks are selected for data extraction forms. We established four quality checks, and each study was analyzed against these checks on a scale of 3 points (fully, partially, no). A study gains 1 point if it fully meets the quality criteria, 0.5 points if it meets the criteria partially, and 0 points if it fails to meet the criteria. The minimum score required to fulfill the criteria is 2.5 out of 4. Additionally, a study will receive 1 point if it is published in an A+ or A-rated conference or in ISI, SCI, or SCIE indexed journals, while studies published in B or C-rated conferences or ESCI, Scopus, DBLP indexed journals will be eligible for 0.5 points. Studies published in other categories will receive 0 points in the quality assessment criteria, as outlined in Table 6.

TABLE 6. Quality assessment criteria.

Criteria#	Description	Score
QC1	Have study clearly defined goals, objectives and methods?	Y|N|P
QC2	Have paper been published in core-ranked conference or indexed journal?	Y|N|P
QC3	Does the research design meet with aims and objectives?	Y|N|P
QC4	Have study explicitly included limitation or future work?	Y|N|P

Data Extraction Data extraction forms were designed to extract data from the selected studies, as illustrated in Figure 3.

The proportional of selected studies is given in the Figure 4.

One hundred and fifty-five studies were selected for data extraction. All necessary features and relevant information were extracted from the selected studies, and quality criteria and inclusion/exclusion checks were applied during this phase. The data extraction form listed items extracted from the studies.

- Title
- Publication venue
- Publication year
- Context of the primary study
- Dataset and its source
- Country
- Application field
- Quality checks
- Evaluation matrices
- Techniques applies
- Limitations
- Future gaps
- Characteristics of the study

Data Extraction Form

Study ID:					
Notes: **Context:**					
Year Of publication					**Notes**
Title					
Publication Venue		☐ Journal	☐ Conference Proceeding		
Authors Name					
Country:		**Eligible**	**Ineligible**	**Unclear**	
Conformance to Inclusion Criteria		☐	☐	☐	
Peer-Reviewed	**Yes** ☐ **No** ☐	**Fully**	**Partially**	**No**	**Score: 2.5**
Quality Criteria	**QC1:** Have study clearly defined goals, objectives and research methods?	☐	☐	☐	
	QC2: Does the research design align with aims and objectives?	☐	☐	☐	
	QC3: Have study explicitly included limitation and future work?	☐	☐	☐	
	QC4: Have paper been published in core-ranked conference or indexed journal?	☐	☐	☐	
Dataset		**Output**			
Application Field					
AI Techniques					
Tools & Techniques (Data Mining)					
Limitations		**Future Research Gaps**			
Contribution Type	**Method/ Technique** ☐	**Tool** ☐	**Framework** ☐	**Model** ☐	**Comparison** ☐
Evaluation Methods					**INCLUDE** ☐ **EXCLUDE** ☐
Reason for Exclusion					

FIGURE 3. Data extraction form.

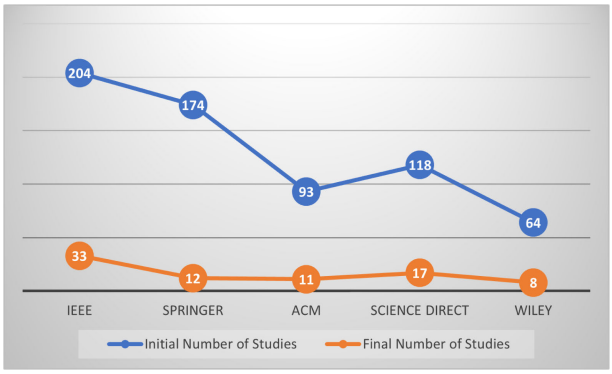

FIGURE 4. Number of initial and final studies.

IV. RESULTS AND DISCUSSION

Taxonomy of Recommender Systems: The taxonomy of AI-based Recommender Systems is shown in Figure 5.

A. RQ1: WHICH DATA SOURCE/DATASET IS MOST FREQUENTLY USED IN RECOMMENDATION SYSTEMS?

The purpose of this research question was to identify the influential data sources in the field of Recommender Systems. Identifying these data sources is crucial as it can guide future researchers in pursuing research in this field. The motivation behind this question was to determine which datasets and data sources have been widely used in Recommender Systems research globally. Table 7 lists the data sources that have garnered the attention of researchers from 2019 to 2024, highlighting that the MovieLens dataset and social media

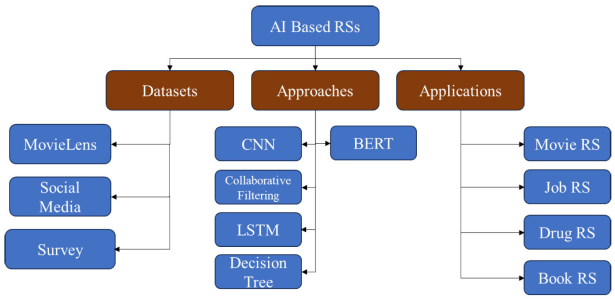

FIGURE 5. Taxonomy of AI-based recommender system.

datasets are the primary data sources in this area of research.

The results from the systematic review indicate that the most widely used data sources are social media and surveys.

1) **Surveys**: Researchers employ various modes of surveys, including mail, in-person interviews, telephone, and Internet-based surveys. The choice of survey mode depends on the data collection requirements. Mail surveys are beneficial for collecting data from large and diverse populations and for achieving a good response rate. Telephone surveys are effective for large samples but can be costly. In-person interviews, while time-consuming and resource-intensive, have been a traditional method for data collection.
2) **Social Media**: With the advent of big social data, the reliance on surveys as a primary data collection method has decreased. Social media platforms now provide vast amounts of data. Online social networks such as Twitter, Facebook, WhatsApp, LinkedIn, Instagram, and YouTube offer extensive data that can be used to identify the professional and behavioral patterns of users.

The frequency of data sources used in different studies has been visualized in Figure 6, which shows that nine studies used the MovieLens dataset, while six studies used social media as a data source.

TABLE 7. Famous dataset and data sources.

Dataset/Data Source	References
MovieLens	[12], [13], [14], [15], [16], [17], [18], [19], [20], [21]
Social Media	[22], [23], [24], [25] , [26], [27]
Survey	[28], [29] , [30] , [31], [32], [33]
Public Repository	[34], [35], [36] , [37] , [38]
BookCrossing dataset and GoodRead dataset	[39], [40]
UCI Machine Learning Repository	[41], [42] , [43]
Website	[44] , [45] , [46] , [47] , [48] , [22], [49] , [50] , [51]
Hospital	[52] , [53] , [54] , [55] , [56]
Education System	[57] , [58] , [59] , [60]
Multi-Objective Recommender System competition on Kaggle	[61]
Amazon Customer Reviews3 and Book Reviews	[62]
Plantão Coronavírus	[63]
Bilibili	[64]
clinical data and rehabilitation data,	[65]
DIGINETICA	[66]
MagNet database	[67]
Automatic identication system (AIS) data	[68]
ALTENcompany	[69]
Newspaper	[70]
Insurance Company	[71]
Other Sources	[72] , [73], [74]

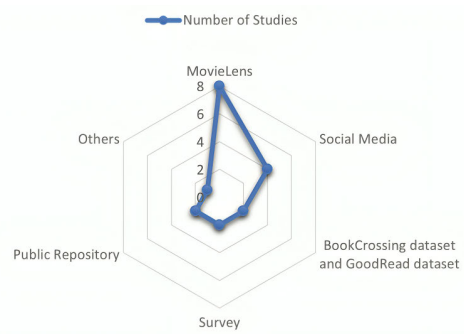

FIGURE 6. Most frequently used datasets/ sources.

B. RQ2: WHAT ARE THE TECHNIQUES AND ALGORITHMS AND EVALUATION METHODS USED IN THE RESEARCH RELATED TO RECOMMENDATION SYSTEMS?

The purpose behind this research question is to identify the approaches that have been used to develop Recommender Systems in various domains. A classified form of these approaches is defined in this section, which will help to compare technical aspects with contextual aspects. From research question 4, it can be seen that Recommender Systems are used in various applications. Each application requires different approaches and techniques to handle effectively.

1) ML AND DL TECHNIQUES

In existing literature different approaches of Machine Learning and Deep Learning are used. The details of technical approaches are given in Table 8 below:

We can see from the Table 8 that seven studies used CNN, five used collaborative filtering, three used LSTM, two used Decision Tree, two used Naive Bayes, two used Multi-layer perceptron. While other techniuqes have been used once in each study.

a: CNN

Convolutional Neural Networks (CNNs) are a class of deep learning models specifically designed for processing grid-like data, such as images and videos [92]. The hallmark of CNNs is their ability to automatically learn hierarchical features through convolutional layers. These layers apply filters across the input data to detect patterns, edges, and textures, progressively capturing more complex structures in deeper layers. Pooling layers further downsample the feature maps, reducing computational complexity while preserving essential information. CNNs excel in image recognition tasks

TABLE 8. Techniques and approaches.

Approaches	Number Of Studies	References
CNN	11	[65], [75], [15], [36], [42], [76], [40], [26], [58], [77], [78]
Collaborative Filtering	10	[13], [14], [22], [79], [71] , [80] , [43], [21] , [81] , [82]
LSTM	5	[63], [83], [67] , [84] , [53]
Decision Tree	4	[52], [41] , [48] , [85]
Naïve Bayes	4	[52], [41] , [43] , [86]
Multilayer Perceptron	3	[16], [17] , [84]
Deep Collaborative Filtering	3	[57] , [87] , [88]
Random Forest	3	[52] , [84] , [86]
Light Gradient Boosting Machine (LightGBM)	2	[61] , [89]
KNN	2	[39] , [43]
Word2vec	2	[44] , [90]
SVM	2	[41] , [84]
Neural Network	2	[28] , [21]
GRU	1	[64]
TextBlob	1	[14]
Content Based Filtering	1	[22]
NLP	1	[68]
Fuzzy Logic	1	[83]
Improved Weighted Bipartite Graph and Slope One	1	[29]
ODROID- XU4 edge computing board	1	[34]
Binary tree and majority voting	1	[34]
Neural collaborative filtering framework	1	[74]
SimRank	1	[70]
TF-IDF	1	[23]
Clustering	1	[91]
Feed Forward DNN	1	[42]
SRDNet	1	[24]
Deep Sequential Learning model architecture	1	[19]

by leveraging shared weights and local connectivity, enabling them to recognize patterns regardless of their position in the input. Widely used in computer vision applications, CNNs have demonstrated exceptional performance in tasks like object detection, image classification, and semantic segmentation, contributing to breakthroughs in fields such as medical imaging, autonomous vehicles, and facial recognition [93].

b: COLLABORATIVE FILTERING

Collaborative Filtering is a powerful technique in recommendation systems that leverages the preferences and behaviors of users to make predictions or suggestions. This approach assumes that users who agreed in the past tend to agree again in the future [94]. There are two main types of collaborative filtering: user-based and item-based. User-based collaborative filtering recommends items based on the preferences of users with similar tastes, while item-based collaborative filtering recommends items that are similar to those a user has liked or interacted with. Collaborative Filtering does not require explicit knowledge of the items being recommended but relies on the collective wisdom of user interactions [80]. It has proven effective in various domains, such as movie recommendations, e-commerce product suggestions, and music playlists, providing personalized experiences to users based on the preferences of like-minded individuals in the user community [95].

c: LSTM

Long Short-Term Memory (LSTM) is a type of recurrent neural network (RNN) architecture designed to address the vanishing gradient problem inherent in traditional RNNs, enabling the model to capture and learn long-range dependencies in sequential data [96]. LSTMs consist of memory cells with self-connected units and incorporate gates that regulate the flow of information, allowing for the retention or deletion of information over extended time intervals. The architecture includes an input gate, a forget gate, and an output gate, each controlling the information flow at different stages. This enables LSTMs to effectively handle sequences of varying lengths and capture dependencies across multiple time steps. LSTMs have demonstrated remarkable success in various applications such as natural language processing, speech recognition, and time-series prediction, making them a fundamental tool in deep learning for sequential data analysis.

LSTM uses logisitc sigmoid function as its activation function given in Equation 1

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{1}$$

In this equation: e is the base of the natural logarithm (Euler's number), approximately equal to 2.71828, while x is the input to the function.

d: DECISION TREE

A Decision Tree is a machine learning algorithm that recursively partitions a dataset into subsets based on the values of input features. Starting with the root node representing the entire dataset, the algorithm selects the best feature to split the data and creates branches corresponding to possible outcomes [97]. This process is applied iteratively to each subset, generating a tree structure with decision nodes representing features, branches representing possible feature values, and leaves containing the final predictions or values. Decision Trees are used for both classification and regression tasks, providing interpretability and versatility. They can handle both numerical and categorical data and are susceptible to overfitting, which can be mitigated through techniques like pruning and ensemble methods. The resulting tree structure facilitates intuitive decision-making and is well-suited for visual representation.

e: NAIVE BAYES

Naive Bayes is a probabilistic machine learning algorithm based on Bayes' theorem, with the "naive" assumption that the features used to describe an instance are conditionally

independent, given the class label. Despite its simplicity and the independence assumption, Naive Bayes has proven to be effective in various applications, especially in text classification and spam filtering. The algorithm is particularly suited for high-dimensional datasets and is computationally efficient [98].

Here's a brief overview of how Naive Bayes works in equation 2 [99]:

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)} \tag{2}$$

where,

- $P(C|X)$ is the posterior probability of class C given evidence X.
- $P(X|C)$ is the likelihood of evidence X given class C.
- $P(C)$ is the prior probability of class C.
- $P(X)$ is the probability of evidence X.

2) EVALUATION MATRICES

Evaluation methods are used to measure the performance of the techniques. Various measures are employed to evaluate these techniques [100]. Table 9 provides an overview of all the evaluation metrics used in this domain. Figure 7 illustrates that the most commonly used measures to assess techniques are precision and recall, followed by Root Mean Square Error (RMSE). Similarly, Mean Square Error (MSE), F1-score, Accuracy, and loss are also utilized to evaluate the performance of the techniques.

C. RQ3: WHAT ARE THE MAIN ARTIFICIAL INTELLIGENCE ALGORITHMS USED IN THIS FIELD?

As generative AI and large language models continue to advance, their roles in Recommender Systems are becoming increasingly significant [102]. In the era of thriving e-commerce and web applications, Recommender Systems (RecSys) have evolved into integral facets of our daily lives, furnishing personalized recommendations tailored to individual user preferences. Despite the substantial progress achieved by Deep Neural Networks (DNNs) in enhancing Recommender Systems by modeling user-item interactions and incorporating textual side information, these DNN-based approaches encounter challenges [103]. These challenges encompass complexities in comprehending users' interests and assimilating textual side information, limitations in adapting to diverse recommendation scenarios, and challenges in providing robust reasoning for their predictions. Simultaneously, the advent of Large Language Models (LLMs), exemplified by ChatGPT and GPT4, has ushered in a paradigm shift in the realms of Natural Language Processing (NLP) and Artificial Intelligence (AI). This shift is attributed to the extraordinary capabilities of LLMs in fundamental language understanding and generation tasks, coupled with their impressive aptitude for generalization and reasoning [104]. Table 10 presents the widely used AI techniques.

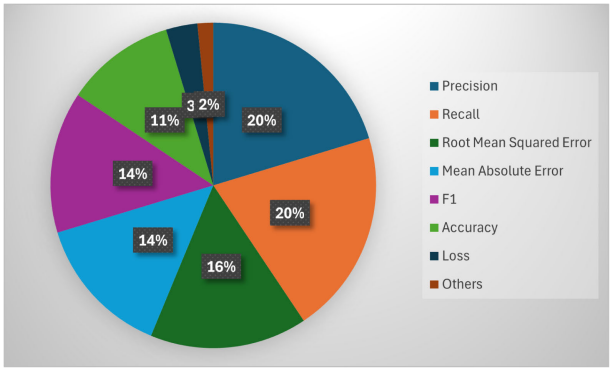

FIGURE 7. Mostly used evaluation matrices.

TABLE 9. Evaluation matrices.

Evaluation Matrices	References
Mean Absolute Error	[39], [62], [12], [65], [67], [28], [15], [16], [24] , [21] , [50], [59], [51] , [45], [47], [37], [35], [43]
Root Mean Squared Error	[39], [14], [13], [12], [15], [16], [34], [34], [18], [24], [40] , [80], [58], [77], [101] , [35]
F1	[63], [44], [75], [34], [69], [74], [36], [91], [42] , [77], [72], [49], [47]
Accuracy	[13], [52], [66], [41], [69], [25], [17] , [37], [47], [21], [43]
Precision	[13], [57], [63], [44], [75], [29], [34], [69], [74], [36], [71], [91], [42] , [30]
Recall	[13], [57], [61], [63], [44], [75], [29], [34], [69], [74], [36], [91], [42], [27]
Response Time	[66]
Loss	[41], [71] , [26]
Bandwidth	[83]
Message Size	[83]
Message Rate	[83]
AUC	[71]
p-value	[76]

1) BERT

Bidirectional Encoder Representations (BERTs) operate on both sides of words, allowing them to understand the context of a word. BERT can perform multiple Natural Language Processing (NLP) tasks simultaneously and is the first bidirectional and deep system for unsupervised learning of NLP tasks [93]. This bidirectional capability enhances the model's ability to capture the context-dependent meaning of words in sentences. Unlike traditional NLP models, BERT weighs the importance of each word in context using attention mechanisms. BERT's capacity to comprehend deep-contextual information has led to its widespread adoption in Recommender Systems [106], [107].

2) ATTENTION MECHANISM

Attention mechanism is a crucial component in modern Natural Language Processing (NLP) and computer vision tasks. During prediction, it enables models to focus on specific parts of the given sequence by weighting each word based on its importance in the context. When generating an output, attention models assign varying levels of importance to different parts of the input, allowing the model to select the

TABLE 10. AI-based approaches.

AI Based Approaches	References	Number of Studies
BERT	[69], [18]	2
Attention Based Model	[62] , [51]	2
Transformer Based Model	[67] , [105]	2
Feed Forward DNN	[42] , [21]	2
Back Propagation (BP) neural network with Attention Mechanism (BPAM).	[12]	1
Triple Attentive Neural Network (TANN)	[66]	1
Adversarial GRU-Attention Matrix Factorization Model	[15]	1
Multi-agent system	[25]	1

relevant portion of the input sequence. In the context of transformers, a popular architecture for sequence-to-sequence tasks, the attention mechanism calculates attention scores for each element in the input sequence. These scores are then used to compute a weighted sum, emphasizing the elements that the model should focus on during processing [108].

a: EXAMPLE OF ATTENTION MECHANISM IN MACHINE TRANSLATION AND IMAGE CAPTIONING

The attention mechanism, a key component in modern AI models, significantly improves the performance of various applications by allowing the model to focus on different parts of the input data based on relevance. **Machine Translation:** In machine translation, such as translating a sentence from English to French, attention mechanisms help the model focus on specific words in the source sentence when generating each word in the target sentence. For instance, when translating the English sentence "The cat sat on the mat" into French, the attention mechanism ensures that when generating the French word for "sat," the model pays particular attention to the English word "sat," and similarly, for "mat," it focuses on "mat." In machine translation, attention mechanisms significantly enhance the model's ability to produce accurate translations. For example, when translating a complex sentence from English to French, attention allows the model to weigh the importance of different words in the source sentence dynamically. Instead of treating all words equally, the model can focus more on the words that are crucial for generating each word in the target language. This results in translations that better capture the nuances and context of the original sentence, improving overall translation quality and coherence. In text summarization, attention mechanisms help the model to focus on the most important sentences or phrases from a document, enabling it to generate concise and coherent summaries. By attending to different parts of the text with varying degrees of importance, the model can effectively capture the essence of the document without losing critical information. This results in summaries that are more relevant and reflective of the key points in the original text, making them more useful and informative for users. **Image Captioning:** Application Example: In image captioning, attention mechanisms enable the model to generate descriptive captions for images by focusing on different regions of the image. For instance, when describing an image of a dog playing with a ball, the attention mechanism allows the model to concentrate on the dog's face when mentioning "dog" and on the ball when mentioning "ball." By dynamically focusing on various parts of the image, the model generates more coherent and contextually accurate captions.

3) INTEGRATION OF AI MODELS WITH RECOMMENDER SYSTEMS

The integration of attention mechanisms and transformers with different types of RSs allows for more dynamic, context-aware, and precise recommendations. These AI models contribute to enhancing the ability of RSs to handle complex data, capture nuanced user preferences, and provide personalized suggestions.

a: COLLABORATIVE FILTERING RSs

Attention Mechanisms: In collaborative filtering, attention mechanisms can enhance user-item interaction modeling by allowing the system to focus on the most relevant past interactions or user preferences. For example, self-attention layers can weigh the importance of different user interactions to improve the prediction of user preferences. **Transformers:** Transformers, with their ability to handle sequential data and capture long-range dependencies, can be used to model user behavior over time. They can process user-item interaction sequences to learn complex patterns and dependencies, which can be beneficial for predicting user preferences in collaborative filtering settings.

b: CONTENT-BASED RSs

Attention Mechanisms: In content-based RSs, attention mechanisms can be employed to highlight the most relevant features of items or user profiles. This approach helps in fine-tuning the focus on specific attributes or content characteristics that align closely with user interests, improving recommendation accuracy. **Transformers:** Transformers can be used to analyze and understand the semantic content of items. By leveraging their capability to process and represent text data through embeddings, transformers can enhance the content-based approach by providing more nuanced and contextually relevant item descriptions and user profiles.

c: HYBRID RSs

Attention Mechanisms: Hybrid RSs, which combine collaborative and content-based approaches, can benefit from attention mechanisms by integrating the strengths of both methods. Attention layers can help in balancing and weighting the contributions of collaborative filtering and content-based features to produce more accurate recommendations. **Transformers:** Transformers can facilitate the fusion of collaborative and content-based features by processing and integrating various data sources and modalities. For example, a transformer model can simultaneously handle user

interaction sequences and item content descriptions, providing a unified and sophisticated recommendation model.

D. RQ4: WHAT ARE THE MAIN APPLICATION FIELDS OF THE RECOMMENDATION SYSTEMS?

In research, an "application area" refers to the specific domain or field where a particular research study or project is focused. It defines the context or industry in which the research findings, methodologies, or technologies are intended to be applied, providing a framework for understanding the practical implications and relevance of the research.

For example, in the context of artificial intelligence or machine learning research, the application area could be in healthcare, finance, natural language processing, computer vision, or any other field where the developed algorithms or models are intended to be used.

Researchers often choose specific application areas based on their expertise, interests, or the societal impact they aim to address. Defining the application area helps researchers tailor their methods and outcomes to the specific challenges and requirements of that domain, allowing for a more targeted and meaningful contribution to the body of knowledge within that particular field. Table 11 presents the main application areas in the field of Recommender Systems.

We can see from Table 11 that nine researchers worked on recommendation in healthcare, while eight primary studies were found to have research on movies recommendations. Academic recommendation and job recommendation systems were proposed by five and four studies respectively. Moreover, three studies for book recommendation and two studies for news recommendation were included in the review.

TABLE 11. Application fields of research done on recommender systems.

Application Field	References	Number of Studies
Health Recommendation	[52], [41], [42], [35], [26], [53], [27] , [56] , [109] , [89]	9
Movie Recommendation	[12], [13], [14], [15], [16], [18], [24], [19]	8
Academics	[57], [76] , [45], [30], [110]	5
Job Recommendation	[44], [69], [91], [25], [86]	5
Book Recommendation	[39], [62], [40]	3
News Recommendation	[79], [70]	2
Tourism Recommendation	[22], [51]	2
Multi Objective Recommendation	[61]	1
Chat Recommendation	[63]	1
Video Recommendation	[64]	1
Cancer Rehabilitation Scheme	[65]	1
Recommendation for Enterprise	[66]	1
Material Recommendation	[67]	1
Port Recommendation	[68]	1
Hyper-parameter Recommendation	[75]	1
Design of Cloud-Edge Data Analysis in Cyber-Physical Systems	[83]	1
Supplement Recommendation	[28]	1
Teacher Recommendation	[29]	1
Energy Efficiency Recommendation	[34]	1
Resource Recommendation	[36]	1
E-Commerce	[43]	1
Feedback Recommendation	[17]	1
Recruitment Recommendation	[23]	1
Insurance Recommendation	[71]	1
Agriculture	[84]	1

Movie recommendation systems, are sophisticated algorithms or applications designed to offer personalized suggestions to users based on their viewing history, preferences, and behavior. These systems employ various techniques, including collaborative filtering, content-based filtering, and machine learning, to analyze user data and identify patterns in movie preferences. By understanding a user's past interactions with the platform and comparing them with similar users, these systems can recommend movies that are likely to align with the individual's tastes. Additionally, content-based filtering considers specific attributes of movies, such as genre, actors, or themes, to provide tailored recommendations. Popularized by streaming services like Netflix, Hulu, and Amazon Prime Video, movie recommendation systems enhance user experience by offering a curated selection of films, fostering user engagement, and helping users discover new content that suits their interests.

Job recommendation systems are specialized applications designed to assist individuals in finding suitable employment opportunities based on their skills, qualifications, and career preferences. Leveraging algorithms and data analytics, these systems analyze the user's professional background, education, work experience, and specific job preferences to generate tailored recommendations. The recommendations may encompass job listings, industries, or positions that align with the user's qualifications and career goals. Job recommendation systems play a crucial role in streamlining the job search process, providing users with targeted and relevant opportunities while helping employers connect with potential candidates efficiently. These systems contribute to the optimization of the recruitment process by facilitating matches that benefit both job seekers and employers, fostering a more effective and personalized approach to the job market.

Healthcare:

- **Personalized Treatment Recommendations**: An in-depth analysis of how recommendation systems are used to tailor treatment plans to individual patients based on their medical history, genetic information, and real-time health data, thereby improving patient outcomes and efficiency in healthcare delivery [35], [111].
- **Clinical Decision Support Systems**: Examination of the role of recommendation systems in assisting healthcare professionals with diagnostic and therapeutic decisions by providing evidence-based suggestions and predicting potential health risks [26].

- **Patient Monitoring and Management**: Discussion on the application of recommendation systems in continuous patient monitoring, such as wearable devices that track vital signs and provide recommendations for lifestyle adjustments or medical interventions [53].
- **Mental Health of Patient** Predicting mental health disorders using recommendation systems represents a significant advancement in personalized healthcare. These systems utilize advanced machine learning algorithms to analyze a wide array of data sources, including electronic health records, wearable device data, social media activity, and patient self-reports. This proactive approach not only helps in early detection and prevention of mental health issues but also enhances the overall quality of mental health care by offering tailored recommendations and continuous monitoring [27].

Academic Applications:

- **Personalized Learning Paths**: Exploration of how recommendation systems are employed to create customized learning experiences for students, helping them achieve better academic performance by suggesting relevant resources, courses, and study schedules based on their learning styles and progress [45].
- **Research Resource Recommendations**: Analysis of systems that recommend relevant research papers, journals, and other academic resources to researchers and students, aiding in efficient literature discovery and staying updated with the latest advancements in their fields [30].
- **Curriculum Development**: Insight into the use of recommendation systems in designing and updating curricula based on student performance data, feedback, and evolving educational trends, ensuring that the curriculum remains relevant and effective [110].

E. RQ5: WHAT IS THE OVERALL PRODUCTIVITY IN THE FIELD OF RECOMMENDATION SYSTEMS?

The purpose of this research question is to have an overview of the research trends in terms of publication venues, most active country, and chronological distribution.

1) ACTIVE COUNTRIES

Active country means that those countries which are active research field. Names of active countries with number of studies is given in the Figure 8.

The Figure 8 shows that the most active country in the field of recommendation system is China. China has published eleven studies in this field while Brazil has published three studies. Two studies have been published in India, USA, Malaysia, Iran and Japan, while all other countries have contributed in this field by publishing one study each.

2) CHRONOLOGICAL DISTRIBUTION

Chronological distribution is shown in Figure 9 to represent the emergence of Recommender Systems. It can be seen from

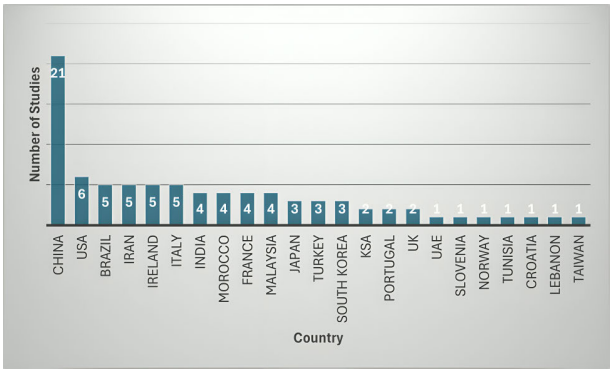

FIGURE 8. Graph showing the number of studies with respect to country.

Figure that the research productivity was high in 2023 and 2019. There was decrement in research of Recommender Systems during year 2020.

3) PUBLICATION VENUES

To identify the publication venues is the important step to examine the research productivity with respect to certain topic. The advantage of identifying the publication venues is that researchers can target the conferences and journals in their future research. Table 12 shows the top journal venues related to Recommender Systems. It can be seen from the table that IEEE Access has published 9 studies in the field of Recommender Systems, while Springer Nature Switzerland AG. has published 6 studies.

The main conferences in this domain are shown in Table 13:

Top conferences in the field of Recommender Systems can be seen from the table 13.

F. RQ6: WHAT ARE THE EXISTING CHALLENGES AND FUTURE RESEARCH OPPORTUNITIES OF THE STUDIES?

After analyzing the results of selected studies, limitations and future research implications have been listed below in Table 14.

1) USED ADVANCED TECHNIQUES

Advanced techniques in Recommender Systems encompass a diverse range of methodologies designed to enhance the accuracy and personalization of recommendations. Among these techniques, deep learning has gained prominence, leveraging neural networks to capture complex patterns and representations within user preferences and item features [42]. Matrix factorization, a method that decomposes user-item interaction matrices into latent factors, continues to be a powerful approach for collaborative filtering. Hybrid models combine multiple recommendation strategies, such as collaborative and content-based filtering, to exploit their respective strengths [40]. Sequential recommendation models consider the temporal aspect of user interactions, incorporating the order and timing of user-item engagements for more context-aware suggestions. Reinforcement learning,

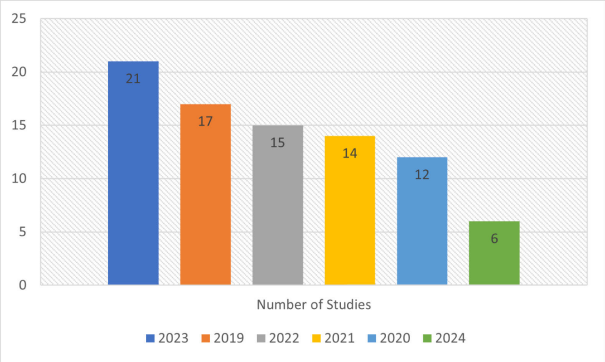

FIGURE 9. Chronological distribution.

an evolving field, enables Recommender Systems to optimize recommendations over time by learning from user feedback [65]. Context-aware recommendation systems integrate additional contextual information, such as location [62], time, or device, to tailor suggestions in diverse settings. These advanced techniques collectively contribute to the evolution of Recommender Systems, striving to provide more accurate, diverse, and personalized recommendations across various domains.

2) ADDITIONAL FEATURES

In the selected primary studies, many studies have the limitation of using few features for recommendations. Hence, these studies have set their future goal to add additional features in Recommender Systems. The features are of many types:

Personal Attributes: Personal attributes are the first kind of contextual information needed for recommendations. They include, age, gender, location, interest and many other similar attributes [66].

Professional attributes can be used to help in recommending Job or career path [69]. They include educational attributes (degree, institution), organizational attributes (job position, job duration, company, company scale, job history, joining and resignation time etc.), specialization attributes (skills, interest).

Behavioral attributes: Behavioral attributes are the third type of contextual information used in the Recommender Systems [91]. They define the attitude, behavior, reputation, moods, sentiments of the users.

Environmental attributes: Environmental attributes also play an important role in recommendations [75]. They provide the information of the users' current socio-economic status, his connections, his sources of income, his social status, his followers [24] etc.

3) USE LARGE SCALE DATASET

Using large-scale dataset is integral in developing the Recommender Systems. For the improvement of Recommender Systems, large scale datasets are necessary for training and accuracy of the advanced algorithms [39]. Large-scale

TABLE 12. Top journals.

Journal	Reference	No. of Studies
IEEE Access	[65], [66], [68], [41], [75], [28], [15], [16], [29], [27], [56]	11
Springer Nature Switzerland AG.	[23], [91], [42], [18], [76], [40]	6
Neural Computing and Applications	[24], [19]	2
Expert Systems With Applications	[70] , [21]	2
Knowledge Based Systems	[51], [77]	2
ACM Transactions on Information Systems	[62]	1
Big Data and Cognitive Computing	[44]	1
International Journal of Advanced Research in Engineering and Technology (IJARET)	[52]	1
Big Data Mining And Analytics	[22]	1
Journal of Web Engineering	[79]	1
IEEE Transactions on Power Electronics	[61]	1
Industrial Electronics Society IEEE	[83]	1
Environmental Challenges	[34]	1
Ain Shams Engineering Journal	[74]	1
Resources Policy	[36]	1
Intelligent Systems in Accounting, Finance and Management	[71]	1
Expert Systems	[25]	1
Mobile Information Systems	[80]	1
IAES International Journal of Artificial Intelligence (IJ-AI)	[43]	1
Journal of AI and Data Mining	[45]	1
Scientometrics	[30]	1
International Journal of Data Science and Analytics	[33]	1
Hong Kong Journal of AI and Medicine	[35]	1
Deep Learning in Personalized Healthcare and Decision Support	[25]	1
IEEE Sensors Journal	[53]	1
International Journal on Digital Libraries	[110]	1
Scientific Programming	[85]	1
Simulation Modelling Practice and Theory	[81]	1
Computational Intelligence and Neuroscience	[109]	1

dataset, containing a large number of users' features help to explore the hidden and intricate patterns hidden in the dataset. Now-a-days, social media platforms and different websites can provides datasets containing a huge numbers of users' features [28]. By leveraging large-scale datasets, recommendation systems can offer more precise and personalized suggestions, enhancing user satisfaction and engagement across diverse domains, from e-commerce and entertainment to education and beyond [76]. However, the management and processing of such extensive datasets also present challenges, requiring robust infrastructure and scalable algorithms to ensure efficiency and performance.

4) ADVANCED AI TECHNIQUES NOT THOROUGHLY EXPLORED

Exploring these advanced AI techniques could provide significant benefits to RSs by enhancing their adaptability, accuracy, and ability to handle complex data. Implementing these methods can lead to more personalized and effective

TABLE 13. Famous conferences.

Conferences	References	Number of Studies
International Conference On Digital Technology in Education	[39]	1
The 3rd International Conference on Advances in Artificial Intelligence (ICAAI 2019)	[13]	1
AICCC: Artificial Intelligence and Cloud Computing Conference	[57]	1
7th International Conference on Big Data Technologies	[67]	1
The first international conference of Web Intelligence and Intelligent Agent Technology	[63]	1
The 3rd International Conference on Machine Learning and Soft Computing (ICMLSC 2019),	[64]	1
The 4th International Conference on E-commerce, E-Business and E-Government	[14]	1
Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI-19)	[12]	1
27th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2023)	[69]	1
Artificial Intelligence and Cognitive Science	[17]	1
Fifth International Conference on Computing Methodologies and Communication (ICCMC 2021)	[84]	1
2nd International Conference on Information Science and Systems	[58]	1
5th EAI International Conference on Smart Objects and Technologies for Social Good	[72]	1
ACM Recommender Systems Challenge	[47]	1
SITA'20: Theories and Applications	[59]	1
The 12th International Conference on Computer Modeling and Simulation	[48]	1
Biannual Conference of the Italian SIGCHI Chapter	[112]	1
SIGIR conference on research and development in information retrieval	[78]	1

TABLE 14. Limitations and future work.

Future Trend	References
Used Advanced Technques	[63], [64], [65], [41], [83], [42], [18], [40], [71], [49], [72], [47], [37] , [78] , [90]
Additional Feature	[61], [14], [66], [75], [69], [91], [24] , [77], [51], [89]
Used Large dataset	[39], [28], [15], [29], [76] , [113], [103]
Improve Accuracy	[13], [16], [34], [23], [80] , [87]
Increase Domain	[36], [17], [68] , [26] , [21] , [86]
Include Other Languages , [58]	[70], [25] ,
Include Geolocation factor	[62]
Improve Problem Defining	[74], [43]

recommendations, ultimately improving user satisfaction and engagement.

a: NEURAL ARCHITECTURE SEARCH (NAS)

Potential Benefits: NAS automates the process of designing neural network architectures, potentially discovering novel and optimized models for RSs. By efficiently searching for the best network structure, NAS can enhance the accuracy and efficiency of recommendation algorithms. It could tailor the architecture to specific RS needs, improving performance in personalized recommendations.

b: GRAPH NEURAL NETWORKS (GNNs)

Potential Benefits: GNNs can model complex relationships in user-item interactions by representing these interactions as graphs. This technique can capture higher-order connections and dependencies, leading to more accurate and nuanced recommendations. GNNs are particularly useful for capturing intricate patterns in large-scale, sparsely connected data.

c: META-LEARNING (FEW-SHOT LEARNING)

Potential Benefits: Meta-learning aims to improve model performance with limited data by learning how to learn from few examples. In RSs, this technique can be beneficial for quickly adapting to new user preferences or item categories with minimal data, enhancing the system's ability to provide relevant recommendations in scenarios with sparse information.

d: SELF-SUPERVISED LEARNING

Potential Benefits: Self-supervised learning uses unlabeled data to pre-train models by generating pseudo-labels or tasks from the data itself. This approach can improve RSs by leveraging large amounts of unstructured or partially labeled data, leading to better feature representations and enhanced recommendation quality.

e: GENERATIVE ADVERSARIAL NETWORKS (GANs)

Potential Benefits: GANs can be used to generate synthetic data or augment existing data, which can help in training RS models with improved robustness and diversity. They can also be employed to create realistic user profiles or simulate user interactions, aiding in the development and testing of recommendation algorithms.

f: CAUSAL INFERENCE MODELS

Potential Benefits: Causal inference models aim to understand the cause-and-effect relationships between user actions and preferences. Integrating these models into RSs can improve the system's ability to make recommendations based on the causal impact of certain features or actions, rather than merely correlational patterns.

V. RECOMMENDER SYSTEMS AND THEIR ETHICAL CHALLENGES

In the past 20 years, RS have been developed focusing mostly on business applications. Even if researchers often adopt a user-centred approach focusing on preference prediction, it is evident that the applications of RS have been driven by online commerce and services, where the emphasis has tended to be on commercial objectives. But RS have a wider impact on users and on society more broadly. After all, they shape

user preferences and guide choices, both individually and socially [114].

A. PRIVACY AND DATA SECURITY

Data Collection: Recommender systems rely heavily on user data to function effectively. This involves collecting vast amounts of personal information, which raises significant privacy concerns. Ensuring that this data is securely stored and anonymized is crucial to protect users' privacy [102]. **Consent:** Users must be adequately informed about what data is being collected, how it will be used, and who will have access to it. Obtaining informed consent is essential to maintaining trust and transparency [115].

B. BIAS AND FAIRNESS

Algorithmic Bias: Recommender systems can inadvertently perpetuate or amplify existing biases present in the training data. This can lead to unfair treatment of certain user groups, reinforcing stereotypes or discriminatory practices [116]. **Equity in Recommendations:** Ensuring that the recommendations are fair and equitable across all user demographics is a critical challenge. Techniques to detect and mitigate bias must be integrated into the development process [117].

C. TRANSPARENCY AND ACCOUNTABILITY

Black Box Nature: Many recommender systems, especially those using deep learning techniques, operate as "black boxes" where the decision-making process is not transparent. Users and stakeholders should have access to understandable explanations of how recommendations are generated [101]. **Accountability:** Clear accountability mechanisms should be established to address any negative consequences arising from the use of recommender systems. This includes having processes in place for users to report issues and for these issues to be resolved effectively [118].

D. MANIPULATION AND AUTONOMY

Manipulative Practices: Recommender systems can be designed to manipulate user behavior in ways that benefit the service provider but may not be in the best interest of the user. For example, promoting content that increases user engagement at the expense of their well-being [119]. **User Autonomy:** Users should have control over their interactions with recommender systems, including the ability to modify or override recommendations and opt-out of data collection [120].

E. IMPACT ON SOCIETY

Filter Bubbles and Echo Chambers: Recommender systems can create filter bubbles by consistently presenting users with content that aligns with their existing preferences and views, limiting their exposure to diverse perspectives and potentially polarizing society [121]. **Cultural and Social Implications:** The widespread use of recommender systems can influence cultural trends and social norms. It is important to consider the broader societal impacts and strive to promote content that supports social good [122]. Addressing these ethical challenges requires a multi-faceted approach, involving the development of robust technical solutions, adherence to regulatory frameworks, and fostering a culture of ethical responsibility among developers and stakeholders. By prioritizing ethical considerations, we can harness the power of recommender systems while minimizing their potential harms.

VI. CONCLUSION

In this study, A SLR was conducted to analyze the recent studies in the field of AI-based Recommender Systems. A thorough search of the major databases was performed to collect the appropriate and high quality research papers published between 2019-2024. Queries were generated for the each databases i.e. IEEE, Science Direct, Springer, ACM and Wiley. We collect 381 primary studies and after performing Prisma flow, only 40 studies were selected for the review. The results of the review showed that the main source of data are surveys and the online networks, the most widely used techniques are CNN, Collaborative filtering, LSTM and Decision tree as well as the most advance AI technique i.e. BERT has been seen to make remarkable progress in AI-based Recommender System field. The review also shows that the main application areas that were explored by RSs are Movie recommendation, Job recommendation, Drug recommendation and Book recommendation. The review puts forward some interesting future research directions as well, that are: to use more advanced techniques, to use additional feature while making recommendation, to use large scale dataset, to increase the language variety etc. in future.

STATEMENTS AND DECLARATIONS

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AUTHORS' CONTRIBUTIONS

All Authors equally contributed to the paper development.

COMPETING INTERESTS

The authors declare no compete of interests.

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